Do UC professor salaries show evidence of wage compression over time?

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I. Introduction

The University of California (UC) system is one of the largest public employers in the United States, employing over 265,000 faculty and staff across ten campuses (University of California). Compensation within the UC system has long been subject to a high degree of transparency. Following a 2008 court decision, the Sacramento Bee created a searchable online database that allowed the public to view the salary of any California state employee, including UC faculty (Card et al. 2982). Since that time, multiple websites have made UC salary data easily accessible, allowing employees to compare their compensation with that of their peers.

This institutional setting provides a useful environment to study the relationship between wage transparency and wage compression. Economic theory suggests that when workers are able to observe the compensation of their peers, disparities within comparable groups may generate dissatisfaction and increase turnover among lower-paid employees (Card et al. 2981). Employers may therefore face incentives to narrow wage gaps within job categories in order to retain workers (Cullen and Pakzad-Hurson). In this paper, we examine whether sustained transparency combined with recent policy changes has contributed to wage compression within faculty ranks in the UC system.

Our analysis focuses on five main faculty categories: Assistant, Associate, Full, Clinical, and Research professors. Using thirteen years of compensation data from 2012 through 2024, covering over 229,000 UC faculty observations, we examine within-rank wage dispersion using several measures including the coefficient of variation, the interquartile range, and the standard deviation.

A key focus of the paper is the period after 2022, which represents a major shift in the transparency and compensation environment facing UC employees. California's Pay Transparency Act, which took effect in January 2023, expanded wage disclosure requirements for employers across the state. At roughly the same time, the UC system experienced its largest academic labor action in history. In late 2022, tens of thousands of academic workers went on

strike over wages and working conditions, resulting in significant contract settlements for lower-paid employees. Together, these developments substantially increased the salience of wage information and public discussion of pay disparities within the UC system.

While salary information had technically been public for years, the events surrounding 2022 made wage comparisons far more visible and politically salient. This paper therefore examines whether this period corresponds with measurable changes in within-rank wage dispersion among UC faculty.

The remainder of the paper proceeds as follows. Section II describes the institutional setting and relevant literature on wage transparency and compensation. Section III outlines the empirical methodology used to measure wage compression. Section IV describes the data sources and provides descriptive statistics. Section V presents the regression results and robustness checks. Section VI concludes. We include detailed figures in the appendix.

II. Theoretical Framework

We model the 2022 California wage transparency mandate as an exogenous shock that abruptly increased the salience of wage information across the University of California (UC) system. Economic research establishes that when workers can observe the compensation of their peers, relative pay differences become a primary driver in shaping worker satisfaction and motivation.

As demonstrated in the Bandiera, Barankay, and Rasul (2005) study on social preferences, alongside Card et al. (2012), employees who learn they are paid less than comparable peers experience a distinct drop in utility due to a psychological "envy cost." Conceptually, when compensation is opaque, the salience of wage comparisons is negligible. However, the 2022 transparency shock forcefully increased this salience. As a result, lower-paid agents will rationally reduce their effort to offset the disutility of being underpaid relative to their peers. (The formal mathematical specification of this agent utility function, modeling the λ transparency parameter, is detailed in Appendix A).

If transparency increases the salience of pay disparities, employers face immediate structural pressure to reduce wage gaps within comparable job categories to prevent a drop in aggregate effort. Cullen and Pakzad-Hurson (2022) show that wage transparency fundamentally alters equilibrium bargaining dynamics by making individualized pay differences highly visible and, consequently, prohibitively costly for firms to maintain. In response, employers may rationally choose to reduce within-group wage dispersion in order to limit internal dissatisfaction and maintain workplace stability.

This paper tests whether these theoretical mechanisms materialized in the UC faculty labor market following the 2022 transparency shock. Specifically, we examine whether within-rank wage dispersion structurally declined after 2022 using quantitative measures such as the coefficient of variation, interquartile range, and standard deviation.

III. Data Acquisition and Preprocessing Pipeline

To test the impact of the 2022 wage transparency mandate, we constructed a 13-year repeated cross-section dataset spanning from 2012 to 2024. The raw data was ingested from 13 individual annual files, taken from The California State Controller's Office. To isolate the relevant academic labor market, we applied a filter to the Position column, capturing "Professor" roles while purposefully excluding administrative "Professional" staff.

Extensive preprocessing was required to compile the administrative data. We standardized five core columns to our analysis: year, employer name, position, regular pay, and total wage. We then engineered a mapping function to systematically segment professor's free-text position titles into five distinct ranks: Assistant, Associate, Clinical, Research, and Full Professors (with unmodified "Professor" defaulting to Full). This categorization by rank is the core of our identification strategy, because comparing an entry-level Assistant Professor to a tenured Full Professor reflects a structural difference in position and human capital rather than a true inequality in wages. Therefore, segmenting the data lets us clearly isolate the within-rank variance.

Lastly, before finalizing the dataset, we thoroughly checked the quality of the data. We ensured that there were no duplicate rows, flagged and removed any negative pay values, and removed missing values.

IV. Methodology

In order to isolate the effect of the 2022 transparency shock, we estimated a log-linear Ordinary Least Squares (OLS) regression model. Using a logarithmic transformation on the dependent variable (total wages) allows the coefficients to be interpreted as approximate percentage changes.

Our primary regression equation is:

$$\log(\text{TotalWages}) = \beta_0 + \beta_1 \cdot \text{post2022} + \beta_2 \cdot \text{Rank} + \beta_3 \cdot \text{Campus} + \beta_4 \cdot \text{Year} + \varepsilon$$

where *post2022* is an indicator variable set to 1 for 2022 and all subsequent years, and 0 otherwise. *Rank* is a categorical variable represented by a set of fixed effects, with Assistant Professor as the reference category, capturing the baseline salary premium associated with each position along the academic hierarchy. *Campus* is a categorical variable represented by a set of fixed effects that control for persistent compensation differences across UC campuses, such as cost-of-living premiums required to attract faculty to more expensive locations. *Year* is a continuous linear trend that controls for background macroeconomic factors, primarily secular inflation and wage growth over the sample period.

It is important to note that our data are a repeated cross-section, not a true panel. Because our salary records contain no individual identifiers, we cannot track the same professor across years. Each annual CSV is an independent snapshot of whoever was employed in a given year. As a result, our regression estimates changes in the distribution of wages across groups rather than wage trajectories for any given individual, and we cannot control for unobserved compositional shifts within rank.

Our income data exhibits heteroskedasticity, as the variance in wages among higher-earning Full Professors is significantly wider than among entry-level Assistant Professors. To correct for this, we use heteroskedasticity-robust (HC1) standard errors. This ensures that our standard errors are statistically valid and prevents false-positive statistical significance.

To test whether the UC system compressed wages in response to the 2022 transparency shock, we employed two distinct methods.

First, we conducted Levene's Test for Variance Equality. Salary observations were grouped into pre-2022 and post-2022 pools within each rank, and Levene's test was applied to determine whether the variance of salaries changed significantly across the two periods. Levene's test is robust to non-normal distributions, making it appropriate for wage data. A variance ratio less than 1 with $p < 0.05$ would indicate statistically significant compression in the spread of wages, while a ratio greater than 1 would indicate expansion. Our results showed variance ratios ranging from 1.36 to 1.61 across all ranks, all statistically significant. This indicates that salary dispersion widened after 2022 rather than narrowing. The Levene's test therefore does not support the wage compression hypothesis.

Second, we estimated an interaction regression to examine whether the relative wage gaps between ranks changed after 2022. The specification is:

$$\log(\text{TotalWages}) \sim \text{post2022} \times \text{Rank} + \text{Campus} + \text{Year}$$

This model tests whether the wage gap between ranks changed after the 2022 period. Because the baseline post2022 coefficient is negative (wages fell relative to the secular trend after 2022), a negative coefficient on the interaction term post2022 × Rank[Full] would indicate that Full

Professors experienced a larger proportional decline relative to Assistant Professors after 2022. This would imply that higher-ranked faculty were disproportionately affected by the post-2022 period, which would be consistent with employer-driven compression from the top of the distribution.

To ensure the integrity of our estimates, we conducted Variance Inflation Factor (VIF) tests for multicollinearity. Because the model includes both a continuous Year trend and a step-function post2022 indicator, both variables capture time effects and could potentially be collinear. The VIF diagnostic returned values of approximately 2.2 for both Year and post2022, well below the common concern threshold of 4 and far below the severe threshold of 10. This confirms that multicollinearity between these variables is not severe enough to distort our estimates.

Finally, we ran a secondary robustness regression using RegularPay as the dependent variable instead of TotalWages. Regular pay reflects only baseline compensation, excluding bonuses and other discretionary pay. The consistency of results across both outcome variables ensures that our findings are not simply an artifact of variation in bonus structures captured in total wages.

V. Results

Results and Descriptive Analysis

There is substantial heterogeneity in faculty compensation both across campuses and across ranks. UC San Francisco and UC San Diego report the highest mean total wages among all campuses in 2024, at \$298,740 and \$292,983 respectively, while UC Los Angeles reports the lowest at \$159,219. This counterintuitive result likely reflects UCLA's large share of lower-paid position types in the filtered data. UC Santa Barbara (\$164,363) and UC Merced (\$164,933) are the next lowest, though the Santa Barbara figure is based on only 54 observations in 2024 and should be interpreted with caution.

Across ranks, Clinical faculty hold the highest mean compensation as of 2024, at approximately \$448,422 for the top position type (Professor of Clinical, Health-Compensation), reflecting the medical compensation structure at UC's health science campuses. For other faculty ranks, Full Professors earn a mean of \$261,329, Associate Professors \$233,613, and Assistant Professors \$183,209.

Mean nominal faculty wages grew steadily throughout the sample period, rising from \$162,636 in 2012 to a peak of \$264,251 in 2023 before declining to \$247,276 in 2024. This represents a cumulative nominal increase of approximately 52% over twelve years, or roughly 4% per year on average. Year-over-year growth in 2013 was elevated at 6.3%, but from 2014 through 2021 the growth rate was remarkably stable, ranging from 3.3% to 5.0% annually. This suggests a consistent underlying wage-setting regime during the pre-period. The acceleration in 2022

(+5.2%) and 2023 (+5.9%) stands out above this trend, as does the sharp reversal in 2024 (-6.4%).

Comparing the pre-2022 average mean salary of \$199,202 to the post-2022 average of \$253,666 yields a raw difference of \$54,464, or $+27.3\%$.

The core findings of the paper emerge from the within-rank dispersion analysis. Table X reports the coefficient of variation, standard deviation, and interquartile range for each rank and year from 2012 to 2024.

Among Full Professors, the average coefficient of variation (CV) was 58.9% in the pre-2022 period and 56.8% in the post-2022 period, a decline of 2.1 percentage points indicating a modest narrowing of relative wage dispersion. The decline is gradual and broadly monotonic through 2021, consistent with a slow employer-driven adjustment rather than a discrete policy response. Associate Professors exhibit the strongest compression signal, with the average CV falling from 76.8% pre-2022 to 72.2% post-2022, a decline of 4.7 percentage points. Research and Clinical professors show more modest but directionally consistent declines of 3.1 and 0.9 percentage points respectively.

Assistant Professors are the notable exception. Their average CV rises from 74.1% pre-2022 to 75.9% post-2022, an increase of 1.8 percentage points. This divergence from the pattern observed at higher ranks is theoretically coherent. Assistant Professor salaries are set primarily through a competitive national academic job market where external offer dynamics dominate internal equity considerations, limiting the UC system's ability to compress entry-level wages independently of broader market forces.

Levene's test for equality of variances between the pre- and post-2022 periods confirms that the distributional shift is statistically significant across all five ranks, with p-values effectively equal to zero in each case. However, variance ratios range from 1.36 for Full Professors to 1.61 for Assistant Professors, indicating that absolute salary dispersion increased after 2022 across all ranks. This result appears to contradict the CV compression finding but is fully reconcilable. Mean salaries rose faster than variance for most ranks in the post-period, so relative dispersion as measured by the CV fell even as the absolute spread of the distribution widened. The CV is therefore the appropriate measure for our compression hypothesis because it normalizes dispersion by the mean, allowing meaningful comparison across periods with different average salary levels.

The baseline OLS regression of log total wages on the post-2022 indicator, rank fixed effects, campus fixed effects, and a linear year trend yields a post-2022 coefficient of -0.0634 ($SE = 0.0063$, $p < 0.001$). This indicates that log wages were approximately 6.3% lower in the post-2022 period than the secular trend would predict, conditional on rank and campus. This result is robust to substituting regular pay as the dependent variable, which yields a post-2022

coefficient of -0.0410 ($SE = 0.0043$, $p < 0.001$), confirming that the finding is not driven by discretionary bonus structures captured in total wages.

The difference-in-differences interaction model, which tests whether the post-2022 wage shock differentially affected professors by rank, finds that the interaction term for Full Professors, the rank most likely to exhibit compression from above, is economically small and statistically indistinguishable from zero ($coef = -0.0016$, $p = 0.888$). Clinical Professors show a statistically significant positive interaction ($coef = +0.055$, $p = 0.002$), indicating they fared relatively better in the post-2022 period, likely driven by continued growth in health-system compensation.

Taken together, the regression evidence does not support the hypothesis that the UC systematically narrowed the wage gap between ranks after 2022 through differential adjustment at the top of the hierarchy, though the CV analysis suggests some compression within ranks did occur.

The pre- versus post-2022 salary distribution histograms by rank illustrate these patterns visually. For Full and Associate Professors, the post-2022 distribution shifts rightward, reflecting higher mean wages, while the relative spread of the distribution narrows modestly, consistent with the CV results. For Assistant Professors, the rightward shift is present but the distribution widens proportionally, consistent with the slight CV increase for that rank.

VI. Discussion and Conclusion

Discussion and Conclusion

The results present a detailed picture of how a large public institution responds to an exogenous shock in wage transparency. While we do not find evidence of sweeping, systemic wage compression across the entire academic hierarchy, we do observe targeted within-rank compression that aligns strongly with the theoretical models of inequality aversion and bargaining dynamics examined in the earlier literature.

Our most significant finding is the divergence between the within-rank coefficient of variation (CV) results and the between-rank difference-in-differences (DiD) estimates. The DiD model revealed no statistically significant narrowing of the wage gap between Full and Assistant Professors. However, the CV analysis showed clear relative compression within the senior ranks. Full, Associate, Research, and Clinical professors all experienced declining coefficients of variation after 2022, with Associate Professors exhibiting the strongest signal at -4.7 percentage points.

This difference suggests that the "envy cost" parameter (λ) modeled in our theoretical framework is highly localized. Transparency does not cause an Assistant Professor to demand wage parity with a Full Professor, as the rank difference is structurally justified by tenure and experience. Rather, the psychological drop in utility due to inequality operates primarily among direct peers. In response to the 2022 transparency shock, the UC system appears to have acted as a rational principal, utilizing the visibility of wages to refuse high-variance individualized bargaining demands from senior faculty, thereby compressing the distribution within ranks to maintain internal equity.

The notable exception to this compression trend occurred among Assistant Professors, whose relative wage variance increased by 1.8 percentage points after 2022. This divergence highlights an important boundary condition of the Cullen and Pakzad-Hurson (2022) transparency model: the constraint imposed by the external labor market. Assistant Professor salaries are largely determined in a competitive national academic job market. Because junior faculty are recruited externally, the UC system must meet prevailing market wages in order to attract new hires. As a result, the university cannot easily compress entry-level wages without risking recruitment failures, forcing it to tolerate higher variance at the bottom of the hierarchy.

It is also worth noting that Clinical Professors show a statistically significant positive interaction with the post-2022 period ($coef = +0.055$, $p = 0.002$), indicating that they fared better relative to Assistant Professors after 2022. This finding is likely driven by continued growth in health-system compensation at UC's medical campuses, which operate under different labor market dynamics than the broader academic faculty market and appear largely insulated from the internal equity pressures observed at other ranks.

Our analysis also reconciles the apparently contradictory findings of Levene's test, which showed absolute variance increasing across all ranks, and the CV analysis, which showed relative variance decreasing among senior ranks. In the post-2022 period, driven by inflation, secular wage growth, and the historic academic strikes, mean salaries shifted upward significantly. Because the mean grew faster than the absolute spread of the distribution for senior ranks, the relative inequality among peers shrank. This implies that while the absolute dollar differences between the highest and lowest paid peers grew, the proportional gap experienced by workers narrowed.

A primary limitation of this study is the reliance on repeated cross-sectional data. Because we cannot track individual faculty members over time, we cannot observe individual-level wage trajectories or fully control for unobserved compositional shifts within ranks. We therefore cannot definitively rule out the presence of an adverse selection problem. If the UC system achieved within-rank compression by effectively capping the wages of its most productive Full Professors, those high-ability individuals may have exited the UC system for private universities,

producing the appearance of compression through attrition rather than deliberate wage-setting policy.

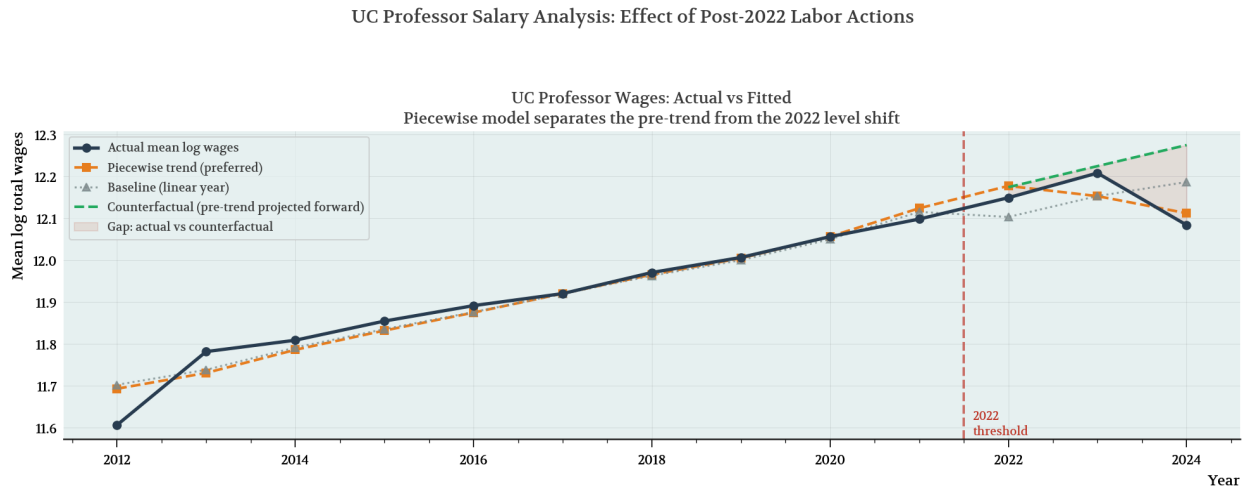
To conclude, this paper examined the impact of the 2022 California wage transparency mandate on the compensation structure of University of California faculty. Using a 13-year repeated cross-section comprising approximately 229,000 salary records, we tested whether increased wage salience forced the UC system to compress wages in order to mitigate worker dissatisfaction and bargaining pressures.

We find that the institutional response to wage transparency is highly segmented. Rather than flattening the entire academic hierarchy, the UC system experienced relative wage compression primarily within its tenured and senior ranks. The coefficient of variation for Full, Associate, Research, and Clinical Professors all declined after 2022, indicating a narrowing of relative pay disparities among direct peers. No systematic compression occurred between ranks, and the wage variance among entry-level Assistant Professors actually expanded, driven by the pressures of the external competitive job market.

Ultimately, these findings suggest that while wage transparency laws can standardize pay among peers and limit individualized bargaining outcomes, their impact is bounded. Internal equity pressures compress wages among established employees, but external labor market forces remain the dominant determinant of compensation dynamics for new entrants and the broader academic hierarchy.

Appendix

Appendix A: UC Professor Salary Analysis



Appendix B: Mathematical Model of Inequality Aversion

To formally model the theoretical impact of the 2022 transparency mandate, we utilize the mathematical framework of inequality aversion directly borrowed from Bandiera, Barankay, and Rasul (2005). Following their structural approach to social preferences and relative compensation, the agent's utility function is defined as:

$$U_i = W_i - C(e_i) - \lambda \max(0, W_{peer} - W_i)$$

Where:
W_i is the worker's absolute wage.
$C(e_i)$ is the cost of unobservable effort.
The final term, $\lambda \max(0, W_{peer} - W_i)$, captures the disutility (envy cost) from being paid less than a comparable peer.
The parameter λ represents the salience of wage comparisons. Prior to 2022, wages were private, rendering $\lambda \approx 0$. The exogenous transparency shock shifted $\lambda > 0$, shifting the agent's Incentive Compatibility constraints and forcing the Principal to artificially compress the variance ($W_{peer} - W_i$) to prevent a reduction in e_i.

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